MIL-R-22097/2F
15 November 1971
SUPERSEDING
MIL-R-22097/2E
4 November 1970

MILITARY SPECIFICATION

RESISTORS, VARIABLE, NONWIRE-WOUND

(ADJUSTMENT TYPE, LEAD-SCREW ACTUATED),

STYLE RJ12

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the detail requirements for style RJ12, adjustment type, lead-screw actuated, nonwire-wound, variable resistors. This style is available in characteristics C and F.

2. APPLICABLE DOCUMENTS

2.1 The following document of the issue in effect on date of invitation for bids or request for proposal forms a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-R-22097 - Resistors, Variable, Nonwire-wound (Adjustment Type), General Specification for.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Requirements. Requirements shall be in accordance with MIL-R-22097, and as specified herein.

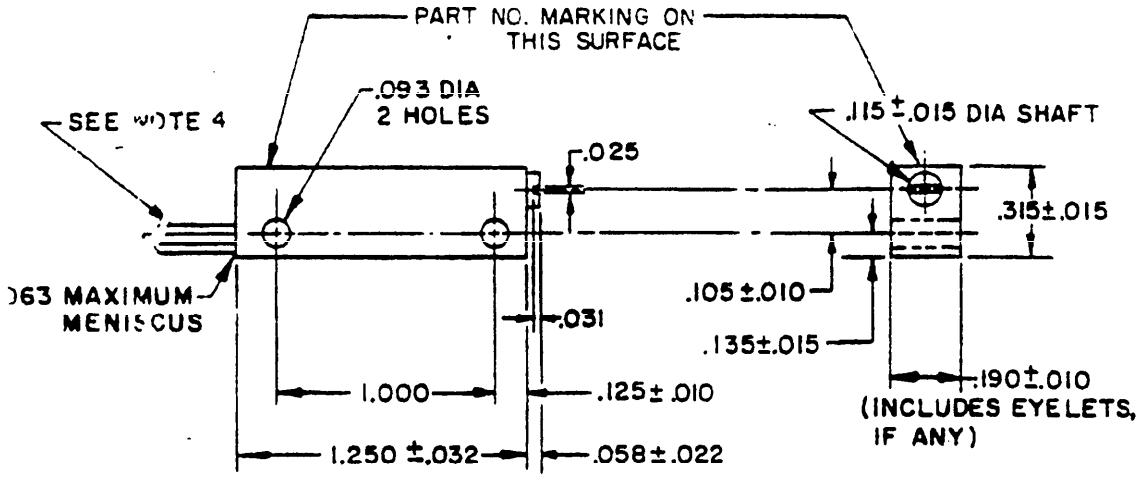
3.2 Design and construction. Resistors shall be of the design, construction, and physical dimensions specified on figure.

3.3 Power rating. The power rating shall be 3/4 watt for characteristics C and F.

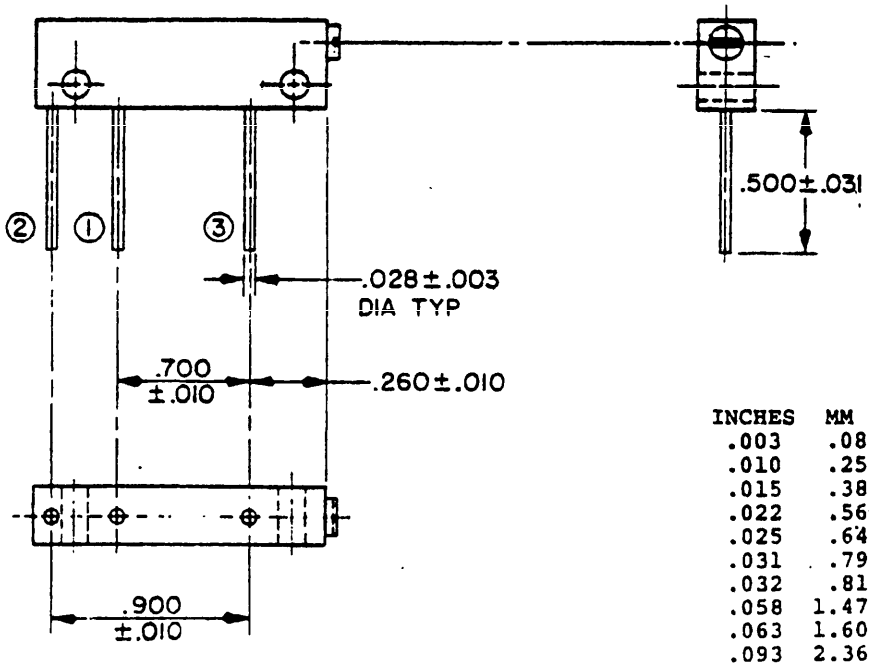
3.4 Terminals. Characteristics C and F are available with L-type, P-type, and Y-type terminals.

3.5 Nominal resistance value and maximum rated ac or dc working voltage. Nominal resistance values and maximum rated ac or dc working voltages shall be as specified in table I.

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FLEXIBLE LEAD TERMINAL TYPE -L

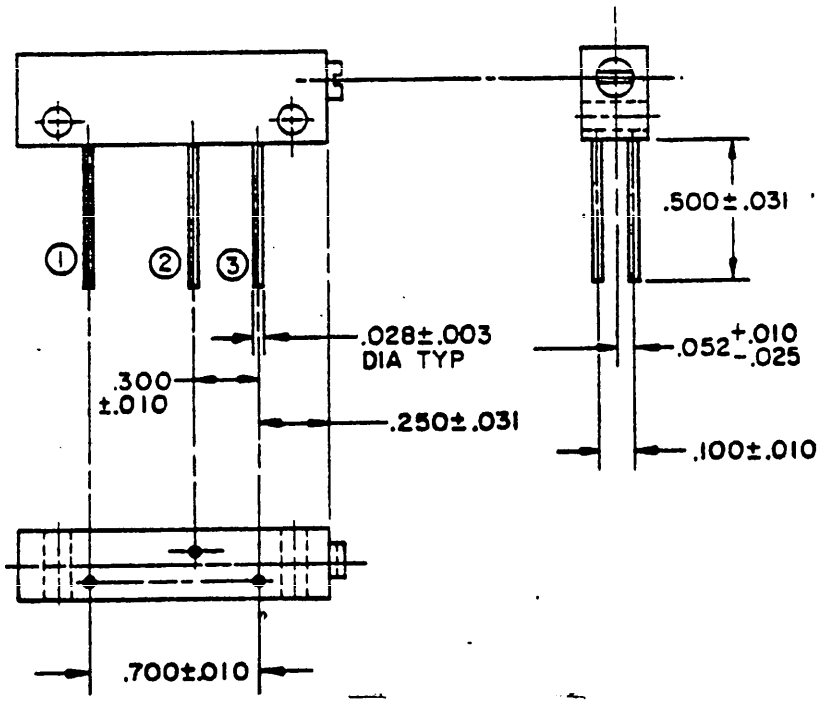


PRINTED-CIRCUIT PIN TERMINAL TYPE-P

INCHES	MM	INCHES	MM
.003	.08	.125	3.18
.010	.25	.135	3.43
.015	.38	.190	4.83
.022	.56	.260	6.60
.025	.64	.315	8.00
.031	.79	.500	12.70
.032	.81	.700	17.78
.058	1.47	.900	22.86
.063	1.60	1.000	25.40
.093	2.36	1.250	31.75
.105	2.67		
.115	2.92		

FIGURE 1. Style RJ12 resistors.

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PRINTED-CIRCUIT
PIN TERMINAL TYPE - Y

INCHES	MM
.003	.08
.010	.25
.025	.64
.028	.71
.031	.79
.052	1.32
.100	2.54
.250	6.35
.300	7.62
.700	17.78

NOTES:
1. Dimensions are in inches.
2. Unless otherwise specified, tolerance is ±.005(.13 mm).
3. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
4. The three leads shall be of stranded wire, AWG size 28 to 30, having a minimum length of 6(152.4 mm); they shall be insulated with polytetrafluoroethylene, stripped .250±.062(6.35±1.57 mm) from the end and color coded.
5. The picturization of the styles above are given as representative of the envelope of the item. Slight deviations from the outline shown, which are contained within the envelope, and do not alter the functional aspects of the device are acceptable.

FIGURE 1. Style RJ12 resistors - Continued.

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TABLE I. Nominal resistance value and maximum rated ac or dc working voltage.

Nominal resistance value	Maximum rated ac or dc working voltage per characteristic
	C, F
Ohms	Volts
10	2.7
20	3.8
50	6.1
100	8.7
200	12.3
500	19.4
1,000	27.4
2,000	38.7
5,000	61.3
10,000	86.7
20,000	122
25,000	136
50,000	194
Megohms	
0.10	274
* 1/ 0.20	300
0.25	300
0.50	300
1.0	300

* 1/ For replacement purpose only, not for design.

3.6 Actual effective electrical travel. Actual effective electrical travel shall be 17 turns minimum, and 27 turns maximum.

3.7 Operating torque. Operating torque shall be a maximum of 8 ounce-inches.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-R-22097, and as specified herein.

4.2 Dielectric withstanding voltage. The magnitude of test voltage shall be 900 volts at atmospheric pressure, and 350 volts at reduced barometric pressure.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with MIL-R-22097.

6. NOTES

6.1 The notes specified in MIL-R-22097 are applicable to this specification.

6.2 Weight. The maximum weight is .00992 pound (4.5 grams).

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* 6.3 Supersession data. This specification supersedes MIL-R-22097/2E, 4 November 1970.

6.3.1 Characteristic B has been deleted. Resistors of characteristic B shall be replaced by characteristic C.

6.3.2 Substitution data. Terminal type P shall not be used for design after 1 November 1969; use terminal type Y.

6.3.3 Characteristic C of this specification shall replace characteristics B and C of MIL-R-22097/1. Characteristic F of this specification shall replace characteristic F of MIL-R-22097/1. Terminal types L and Y of this specification shall replace terminal types L and P respectively, of MIL-R-22077/1.

6.4 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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User activities:
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5. PROBLEM AREAS

a. Paragraph Number and Wording:

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